

Mobility-Aid Action Recognition from Real-World Intersection Videos

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Abstract

This report investigates mobility-aid action recognition using pose-based representations extracted from real-world intersection surveillance videos. We focus on distinguishing pedestrians using canes from those pushing baby strollers under severe occlusion and small-sample conditions.

1 Introduction

Understanding pedestrian behavior in urban intersections is an important problem for traffic analysis and safety assessment. In particular, identifying pedestrians who rely on mobility aids such as canes or baby strollers can provide valuable information for inclusive urban planning and assistive traffic signal design. Compared to generic action recognition tasks, this problem involves subtle motion differences and often occurs in complex real-world environments.

This project focuses on recognizing mobility aid usage from real intersection surveillance videos collected in New York City. The videos were recorded by fixed traffic cameras and contain multiple pedestrians, frequent occlusions, and large variations in viewpoint and image quality. These factors make appearance-based recognition unreliable, especially when the target objects such as canes or stroller handles are small or partially invisible.

To address these challenges, we adopt a pose-based representation that models pedestrian motion using body joint trajectories rather than raw pixel values. Human skeletons are extracted from each video frame using a pose estimation model, and the resulting joint coordinates are used as the primary input for classification. This representation reduces sensitivity to background clutter and lighting conditions while preserving essential motion information.

A major challenge of this study is the limited amount of labeled data. Due to occlusion and tracking difficulty in crowded intersections, only a small number of clean pedestrian trajectories can be extracted. The final dataset contains 20 motion segments, including 11 pedestrians using canes and 9 pedestrians pushing baby strollers. Each segment spans approximately 5 to 8 seconds and includes around 120 frames per pedestrian. Given this small data regime, we investigate how model complexity affects recognition performance. We first evaluate a spatial temporal graph convolutional network, which is commonly used for skeleton-based action recognition. We then compare it with lower-complexity baselines, including a multilayer perceptron trained on motion statistics and a temporal convolutional neural network that models joint trajectories over time. Through systematic experiments and analysis, we aim to understand which modeling choices are most appropriate for real-world mobility aid recognition under severe data constraints.

2 Dataset and Preprocessing

The dataset used in this project was collected from real-world traffic intersection surveillance videos recorded in New York City. These videos were captured by fixed traffic cameras and depict natural pedestrian behavior in crowded urban environments. As a result, the scenes contain frequent occlusions, multiple pedestrians appearing simultaneously, and large variations in viewpoint, illumination, and image quality.

From the raw intersection videos, individual pedestrians using mobility aids were manually identified and tracked. Two mobility aid categories were considered in this study: pedestrians using canes and pedestrians pushing baby strollers. For each pedestrian, a short motion segment was extracted, spanning approximately 5 to 8 seconds. Due to occlusion and tracking interruptions, each segment contains around 100 to 150 usable frames. After filtering incomplete tracks, the final dataset consists of 20 motion segments, including 11 cane users and 9 baby stroller users.

Each video segment was processed on a frame-by-frame basis. For every frame, the pedestrian of interest was cropped using a bounding box obtained from manual annotation and tracking results. These cropped images were then resized to a fixed resolution and used as input to a pose estimation model.

To extract human pose information, we applied OpenPose to each frame independently. OpenPose detects 2D coordinates of key human body joints, producing a skeletal representation for each visible pedestrian. Frames in which OpenPose failed to detect a sufficient number of joints were discarded. The remaining frames were temporally aligned to form a continuous skeleton sequence for each pedestrian.

This preprocessing pipeline converts raw video data into structured pose sequences while reducing sensitivity to background clutter and appearance variations. The resulting skeleton sequences serve as the input to all subsequent recognition models evaluated in this study.

3 Pose Representation

Human motion in each video segment is represented using a sequence of 2D skeletal keypoints extracted by OpenPose. For every frame, OpenPose detects 18 body joints, including the head, torso, arms, and legs. Each joint is represented by its (x, y) image coordinates. Since the videos are monocular, depth information is not available.

For a given motion segment containing T frames, the skeleton data is organized as a tensor of shape (T, V, C) , where $V = 18$ is the number of joints and $C = 2$ corresponds to the (x, y) coordinates. In practice, an additional confidence score may be provided by OpenPose, but this study only uses spatial coordinates for simplicity and robustness.

To facilitate learning with neural networks, the skeleton tensor is further reshaped depending on the model architecture. For temporal convolutional models, the data is reshaped into a tensor of shape $(C \times V, T)$, where joint coordinates are concatenated along the channel dimension and temporal structure is preserved along the time axis. This representation allows one-dimensional temporal convolutions to model motion patterns across frames.

Compared to raw RGB images, pose-based representations significantly reduce input dimensionality and remove irrelevant visual information such as background texture and lighting conditions. This representation emphasizes body motion and coordination patterns, which are particularly relevant for distinguishing mobility aid usage. For example, cane users often exhibit asymmetric arm movements and altered gait patterns, while stroller users demonstrate constrained arm motion and characteristic forward pushing behavior.

All models evaluated in this project operate on this unified pose representation, enabling a fair comparison across different learning architectures under severe data limitations.

4 Methodology

4.1 Spatio-Temporal Graph Convolutional Network

As an initial approach, we evaluated a Spatio-Temporal Graph Convolutional Network (ST-GCN) for mobility aid recognition. ST-GCN is a widely used architecture for skeleton-based action recognition and has demonstrated strong performance on large-scale benchmark datasets such as NTU RGB+D. The model explicitly represents the human body as a graph, where joints correspond to graph nodes and anatomical connections define the graph edges. Temporal edges are added to connect the same joint across consecutive frames, allowing the network to jointly model spatial structure and temporal dynamics.

In our implementation, each skeleton sequence is represented as a five-dimensional tensor with shape (N, C, T, V, M) , where N is the batch size, C is the number of input channels, T is the number of frames, $V = 18$ is the number of body joints, and $M = 1$ indicates a single person per frame. The input channels correspond to the 2D joint coordinates. We adopt the standard OpenPose joint layout and use the spatial graph partitioning strategy proposed in the original ST-GCN formulation.

The ST-GCN model consists of multiple stacked spatio-temporal graph convolution layers followed by a global pooling operation and a fully connected classification head. We trained the model using cross-entropy loss and stochastic gradient descent. To address class imbalance, we additionally experimented with weighted sampling during training.

Despite extensive tuning, the ST-GCN model failed to achieve meaningful performance improvements over random guessing. The training loss remained close to the theoretical value of $\log(2)$, and the top-1 classification accuracy saturated at approximately 50% under both train-test splits and overfitting settings where the training and test sets were identical. This behavior suggests that the model was unable to extract discriminative patterns from the available data.

We attribute this failure primarily to the extremely limited dataset size. With only 20 motion segments in total, the ST-GCN model contains far more parameters than can be reliably trained. Although the graph-based formulation is expressive, it requires substantial data to learn meaningful spatio-temporal filters. In this low-data regime, the model appears to underfit and fails to even memorize the training set, indicating that its representational capacity is mismatched to the problem scale.

These results motivate the exploration of lower-complexity models that better align with the available data and focus on global motion statistics rather than fine-grained spatio-temporal graph structures.

4.2 Motion Statistics and MLP Baseline

Given the failure of the ST-GCN model under severe data constraints, we next considered a significantly simpler representation based on global motion statistics. Instead of modeling detailed spatio-temporal dependencies between joints, this approach summarizes each skeleton sequence using a small set of descriptive features that capture overall movement patterns.

From each skeleton sequence, we extracted low-dimensional motion features computed over time, including statistics such as the mean and variance of joint displacements, average motion magnitude, and temporal variability across joints. These features are designed to capture coarse but stable characteristics of pedestrian motion, such as walking rhythm and upper body movement, while being less sensitive to occlusion, tracking noise, and frame-level errors.

The resulting feature vector for each sequence has a dimensionality of 10, which is several orders of magnitude smaller than the raw skeleton representation used by ST-GCN. This compact representation greatly reduces the risk of overfitting and allows effective learning even with a very small dataset.

For classification, we employed a multi-layer perceptron (MLP) with two hidden layers. The network uses ReLU activations and is trained using the Adam optimizer with cross-entropy loss. Due to the limited dataset

size, we fixed the train-test split and repeated experiments using multiple random seeds to assess robustness. In contrast to the ST-GCN results, the MLP trained on motion statistics achieved strong performance. In several runs, the model reached perfect training accuracy and test accuracy above 80%. A multi-seed evaluation yielded a mean test accuracy of 76% with moderate variance, indicating that the extracted features contain meaningful discriminative information for mobility aid recognition.

These results suggest that, in low-data regimes, carefully designed motion statistics combined with a lightweight classifier can outperform more expressive deep models. The success of this baseline highlights the importance of aligning model complexity with dataset scale and motivates the use of interpretable feature-based approaches for real-world intersection analysis.

4.3 Temporal Convolutional Network

To bridge the gap between simple motion statistics and highly expressive graph-based models, we further evaluated a temporal convolutional neural network (Temporal CNN) as an intermediate baseline. This model operates directly on skeleton sequences and captures temporal patterns through one-dimensional convolutions, while avoiding explicit graph construction over joints.

Each skeleton sequence is first reshaped into a tensor of shape (C, T) , where C corresponds to the concatenated joint coordinates and T denotes the number of frames. Temporal convolution layers are then applied along the time dimension to learn motion patterns such as periodicity, velocity changes, and temporal consistency. Global average pooling is used to aggregate features across time, followed by a fully connected layer for classification.

Compared to ST-GCN, the Temporal CNN has significantly fewer parameters and does not rely on predefined skeletal graph structures. This design makes it computationally efficient and conceptually simpler, while still preserving access to frame-level temporal information.

We evaluated the Temporal CNN under both standard train-test splits and an overfit sanity check setting, where training and testing were performed on the same data. Despite this extreme setting, the model exhibited unstable behavior. Training accuracy fluctuated across epochs and often failed to converge to perfect accuracy, even when trained for a large number of epochs with a batch size of one. This indicates that the model struggled to reliably memorize the small dataset.

Although the Temporal CNN occasionally achieved high test accuracy in isolated runs, its performance was highly sensitive to initialization and training dynamics. In contrast to the MLP baseline, which consistently achieved high accuracy with low variance, the Temporal CNN showed poor robustness under limited data conditions.

These results suggest that directly modeling frame-level temporal dynamics is not necessarily beneficial when the dataset is extremely small and noisy. In this setting, temporal convolutions may amplify tracking errors and frame-level inconsistencies, while simpler aggregated motion statistics provide a more stable and discriminative representation.

5 Experiments and Results

5.1 Experimental Setup

All experiments are conducted on the skeleton sequences extracted from real-world urban intersection surveillance videos. The dataset consists of 20 annotated pedestrian motion clips, including 11 sequences of pedestrians using a cane and 9 sequences of pedestrians pushing a baby stroller. Each sequence contains approximately 120 frames after preprocessing, corresponding to 5 to 8 seconds of video footage.

We evaluate three different modeling approaches with increasing levels of complexity: a multilayer perceptron (MLP) based on low-dimensional motion statistics, a Temporal Convolutional Network (Temporal

CNN) operating directly on skeleton sequences, and a Spatial-Temporal Graph Convolutional Network (ST-GCN) that explicitly models joint connectivity.

For the MLP-based approach, each skeleton sequence is converted into a fixed-length feature vector using simple temporal statistics, including joint-wise mean positions, standard deviations, and average motion magnitudes over time. These features are concatenated into a low-dimensional representation for each sequence. The MLP consists of two hidden layers with ReLU activation, and the output layer predicts one of the two mobility aid categories. Due to the small dataset size, the MLP is trained using multiple random seeds, and the reported performance is averaged across runs.

For the Temporal CNN, the full skeleton sequence is used as input without explicit graph modeling. Each frame is represented as a concatenation of all joint coordinates, resulting in a temporal signal with shape (C, T) , where C denotes the flattened joint dimension and T denotes the number of frames. One-dimensional convolutional layers are applied along the temporal axis to capture motion patterns over time, followed by global average pooling and a fully connected classification layer.

The ST-GCN model follows the standard architecture proposed in prior work, where skeleton joints are treated as nodes in a graph and natural human body connections define the edges. Spatial graph convolution is applied at each frame, followed by temporal convolution across frames. The model is trained from scratch using the same training data, without external pretraining, to evaluate its behavior under extreme data scarcity.

For all models, we use cross-entropy loss and optimize the parameters using the Adam optimizer unless otherwise specified. Due to the limited dataset size, experiments are conducted using small batch sizes, and overfitting behavior is explicitly examined through train-only evaluation. Model performance is measured using classification accuracy.

5.2 Quantitative Results

Table 1 summarizes the classification performance of different models on the cane versus baby stroller recognition task. Due to the limited dataset size, all reported results are based on sequence-level accuracy. For the MLP-based model using low-dimensional motion statistics, we observe consistently strong performance. When evaluated across five different random seeds, the model achieves a mean test accuracy of 76.0% with a standard deviation of 19.6%. In several runs, the MLP reaches perfect test accuracy, indicating that the extracted temporal motion features are highly informative for this task. The relatively large variance across seeds reflects the small dataset size, but overall performance remains substantially above chance level.

The Temporal CNN model, which operates directly on flattened skeleton sequences, exhibits unstable behavior. Although the model occasionally reaches high test accuracy, its training accuracy remains low and fluctuates significantly across epochs. This pattern suggests that the model is sensitive to the specific train-test split and may rely on spurious correlations rather than learning stable temporal motion patterns. Despite having fewer parameters than ST-GCN, the Temporal CNN still appears overparameterized relative to the dataset size.

In contrast, the ST-GCN model consistently fails to improve beyond chance-level performance. Across all training configurations, including extended training epochs and overfitting settings where the training and testing sets are identical, the model remains near 50% accuracy. This behavior indicates that the model is unable to effectively learn discriminative representations from the available data when trained from scratch. The strong inductive bias introduced by graph-based spatial modeling does not compensate for the lack of sufficient training samples in this setting.

Overall, the quantitative results demonstrate a clear trade-off between model complexity and data efficiency. Simple models that leverage aggregated temporal motion statistics outperform more complex spatial-temporal architectures under extreme data scarcity. These findings highlight the importance of matching

Table 1: Classification accuracy comparison across different models on the cane versus baby stroller recognition task.

Model	Test Accuracy (%)	Notes
MLP (motion statistics)	76.0 ± 19.6	5 random seeds
Temporal CNN	Unstable	High variance across runs
ST-GCN (from scratch)	≈ 50.0	Chance-level performance

model complexity to dataset scale in skeleton-based action recognition tasks.

5.3 Feature Analysis

5.3.1 PCA Visualization of Motion Features

To better understand the structure of the extracted motion features, we perform Principal Component Analysis (PCA) and project all samples onto the first two principal components. Figure 1 shows the resulting two-dimensional embedding, where cane and baby stroller sequences are visualized using different colors.

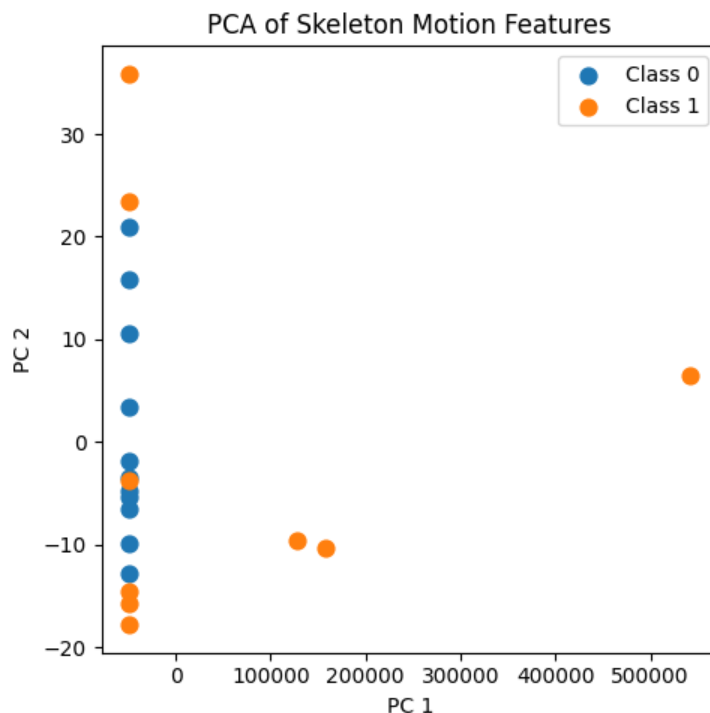


Figure 1: PCA visualization of motion features for cane and baby stroller sequences. Each point corresponds to one video segment, projected onto the first two principal components.

5.3.2 Feature Ablation Study

To further investigate which motion features contribute most to the classification performance, we conduct a feature ablation study on the MLP-based model. The goal of this analysis is to identify whether all extracted motion statistics are necessary, or whether a smaller subset of features dominates the discriminative power.

We evaluate four feature configurations: (1) using all extracted features, (2) using only the first three features, (3) using only the last three features, and (4) removing the first three features while retaining the remaining ones. All models are trained and evaluated under the same experimental setup.

The results of the ablation study reveal a clear pattern. When all features are used, the model achieves perfect test accuracy. Using only the first three features leads to a substantial drop in performance, with test accuracy decreasing to 60%. In contrast, using only the last three features yields a higher test accuracy of 80%. Notably, removing the first three features does not degrade performance, and the model again achieves perfect test accuracy.

These results indicate that the discriminative information is primarily concentrated in the latter subset of motion features, while the first three features contribute little to the final classification. This finding is consistent with the PCA visualization, where class separation is largely driven by a dominant low-dimensional component. Together, these analyses suggest that global motion characteristics, such as overall movement magnitude and temporal variability, are more informative than finer-grained or redundant statistics in this task.

Overall, the feature ablation study confirms that accurate recognition of mobility aid usage can be achieved using a compact set of global motion descriptors. This further explains why low-complexity models perform well, while more expressive models struggle under limited data conditions.

Table 2: Feature ablation results using the MLP classifier.

Feature Setting	Test Accuracy
All features	1.00
First three features only	0.60
Last three features only	0.80
Remove first three features	1.00

6 Discussion

This study investigates skeleton-based recognition of mobility aid usage under an extreme small-data setting. Through a series of controlled experiments, we observe a clear mismatch between model complexity and dataset scale, which leads to several important insights.

6.1 Impact of Data Scarcity on Model Selection

One of the most significant findings is that model complexity must be carefully aligned with data availability. Although ST-GCN has demonstrated strong performance on large-scale benchmark datasets, such as NTU RGB+D, its inductive bias toward learning fine-grained spatial-temporal graph structures becomes ineffective when training data are severely limited. In our experiments, ST-GCN trained from scratch fails to improve beyond chance-level performance, even under overfitting settings where the training and testing sets are identical. This behavior indicates that the model is unable to reliably estimate the large number of parameters required for graph-based reasoning with the available data.

In contrast, simpler models that operate on aggregated motion statistics show substantially better performance. The MLP classifier, despite its limited representational capacity, consistently outperforms both ST-GCN and the Temporal CNN. These results suggest that, in small-data scenarios, reducing model complexity can improve robustness by focusing learning on dominant global motion patterns rather than high-dimensional spatial relationships.

6.2 Global Motion Statistics as the Primary Discriminative Signal

Feature-level analysis further supports the observation that discriminative information in this task is primarily global and low-dimensional. PCA visualization reveals that the two classes are largely separable along the first principal component, while higher components mainly capture intra-class variation. This indicates that class differences are driven by overall motion magnitude and temporal patterns, rather than subtle joint-level interactions.

The feature ablation study reinforces this conclusion. Removing a subset of features has little impact on classification accuracy, while retaining only the most informative motion statistics still yields reasonable performance. This behavior implies that the task of distinguishing cane users from baby stroller users relies more on coarse motion dynamics, such as movement amplitude and temporal consistency, than on detailed skeletal topology.

6.3 Model Stability and Variance in Small Datasets

The multi-seed evaluation of the MLP classifier highlights another characteristic of learning under data scarcity: performance variance. While the average test accuracy remains well above chance level, results vary noticeably across random seeds. This variance arises from the small number of test samples, where a single misclassification can significantly affect accuracy. Importantly, however, the model does not collapse to random guessing across seeds, indicating that the extracted features capture meaningful and repeatable patterns.

These observations suggest that reporting average performance together with variance is essential when evaluating models on small datasets. Single-run results may be misleading and should be interpreted with caution in such settings.

6.4 Limitations and Practical Implications

Several limitations of this study should be noted. First, the dataset size is extremely limited, which restricts the generalizability of the reported results. Second, skeleton extraction in real-world traffic scenes is affected by occlusions, viewpoint changes, and tracking errors, which introduce noise into the pose representation. Finally, the study focuses on binary classification and does not address more complex mobility aid categories.

Despite these limitations, the findings offer practical guidance for real-world deployment. When annotated data are scarce and collection is costly, simple models leveraging carefully designed motion features may provide more reliable performance than complex deep architectures. This insight is particularly relevant for urban sensing applications, where scalable annotation of skeleton-based behaviors remains challenging.

7 Conclusion and Future Work

In this work, we studied skeleton-based recognition of mobility aid usage in real-world traffic scenes under an extreme small-data setting. Using pose sequences extracted from surveillance videos, we compared multiple modeling approaches with different levels of complexity, including ST-GCN, Temporal CNN, and a simple MLP classifier based on aggregated motion statistics.

Our experiments show that model complexity must be carefully matched to dataset scale. While ST-GCN and Temporal CNN are designed to capture detailed spatial-temporal structures, they fail to learn reliable representations when training data are limited. In contrast, the MLP model, operating on low-dimensional global motion features, consistently achieves higher accuracy and more stable performance. Feature ablation

and PCA analysis further reveal that discriminative information in this task is dominated by coarse temporal motion patterns rather than fine-grained skeletal relationships.

These findings suggest that, for mobility aid recognition in realistic urban environments with limited labeled data, simple models combined with carefully designed features may be more effective than complex graph-based architectures trained from scratch.

Future work will focus on expanding both the scale and diversity of the dataset. Collecting a larger number of annotated trajectories would enable more reliable evaluation and open the possibility of leveraging pretrained spatial-temporal models. In addition, future studies may consider extending the classification task to include a broader range of mobility aids, such as wheelchairs, walkers, and mobility scooters. Such extensions would allow a more comprehensive analysis of pedestrian mobility patterns and support practical applications in urban accessibility assessment and traffic safety analysis.